

Assignment 5: Agent Evaluation Dataset

Trace — Personal Productivity and Task Execution Agent

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This document contains 36 test cases across five categories, used to evaluate the Trace agent's tool-selection, execution, and approval behavior. For every case, the expected tool, arguments, approval requirement, and outcome are documented below. The **Actual outcome**, **Pass/Fail**, and **Notes** fields are left blank to be filled in after running each request against the live application (see instructions below), and the Evaluation Metrics table at the end should be completed once all cases have been run.

How to run this evaluation: (1) Start the app locally. (2) Type each “User request” into the chat exactly as written. (3) Check the Execution Logs view to see which tool was actually called, with what arguments, whether approval was requested, and the final result. (4) Compare this to the Expected columns and record the Actual outcome and Pass/Fail for that case. (5) After all cases are run, compute each metric in the Evaluation Metrics table from the recorded results.

Direct Response Cases

Requests that should not trigger any tool • minimum 5 required • provided: 6

Case 1 DIRECT	
User request	What's the difference between High and Critical priority?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent explains the priority levels in plain language. No tool is called.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 2 DIRECT	
User request	What statuses can a task have in this app?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent lists Pending, In Progress, Blocked, Completed, Cancelled. No tool call.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 3 DIRECT	
User request	How does the approval process work here?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent explains that write/irreversible actions show an approval card before running. No tool call.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 4 DIRECT	
User request	What can you help me with?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent summarizes its capabilities (tasks, notes, planning, reminders). No tool call.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 5 DIRECT	
User request	What's the difference between a task and a note in Trace?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent explains tasks track actionable work with status/priority, notes store free-form information. No tool call.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 6 DIRECT	
User request	What does "Suggested by Trace" mean on a task card?
Expected tool(s)	None — direct answer
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent explains this label marks tasks the agent proposed (e.g. from meeting notes) rather than ones the user typed manually. No tool call.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Single-Tool Cases

Requests requiring exactly one tool • minimum 8 required • provided: 9

Case 1 SINGLE-TOOL	
User request	Create a task to review the analytics dashboard specs, due July 22, high priority.
Expected tool(s)	create_task
Expected arguments	<code>{"title":"Review analytics dashboard specs","priority":"High","due_date":"2026-07-22"}</code>
Approval requirement	No (single task creation)
Expected outcome	Task is created immediately; confirmation with a new task ID is returned.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 2 SINGLE-TOOL	
User request	Show me all my critical priority tasks.
Expected tool(s)	list_tasks
Expected arguments	<code>{"priority":"Critical"}</code>
Approval requirement	No
Expected outcome	Returns the list of tasks with priority = Critical and a total count.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 3 SINGLE-TOOL	
User request	List tasks that are due before July 26.
Expected tool(s)	list_tasks
Expected arguments	<code>{"due_before":"2026-07-26"}</code>
Approval requirement	No
Expected outcome	Returns matching tasks and total count.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 4 SINGLE-TOOL	
User request	Mark task TSK-1021 as complete.
Expected tool(s)	complete_task
Expected arguments	<code>{"task_id": "TSK-1021"}</code>
Approval requirement	Yes
Expected outcome	Approval card is shown with the task ID and effect; only marked Completed after the user approves.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 5 SINGLE-TOOL	
User request	Search my notes for anything about onboarding feedback.
Expected tool(s)	search_notes
Expected arguments	<code>{"query": "onboarding feedback"}</code>
Approval requirement	No
Expected outcome	Returns ranked matching notes with scores.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 6 SINGLE-TOOL	
User request	Save a note titled "Vendor call summary" under category Meetings with this content: [content].
Expected tool(s)	save_note
Expected arguments	<code>{"title": "Vendor call summary", "category": "Meetings", "content": "..."} </code>
Approval requirement	No
Expected outcome	Note is saved immediately; a new note ID is returned.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 7 SINGLE-TOOL	
User request	Set a reminder for tomorrow at 9am to follow up on the investor deck.
Expected tool(s)	create_reminder
Expected arguments	<code>{"message":"Follow up on the investor deck","remind_at":"<tomorrow>T09:00:00"}</code>
Approval requirement	Yes
Expected outcome	Approval card is shown; reminder is scheduled only after approval.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 8 SINGLE-TOOL	
User request	Draft a follow-up email to the investor about the revised deck.
Expected tool(s)	draft_followup_email
Expected arguments	<code>{"recipient_context":"Investor","purpose":"Follow up on the revised deck"}</code>
Approval requirement	No for drafting; Yes before it is treated as sent
Expected outcome	A subject/body draft is returned for review. Nothing is sent automatically.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 9 SINGLE-TOOL	
User request	How many tasks do I have that are still pending?
Expected tool(s)	list_tasks
Expected arguments	<code>{"status":"Pending"}</code>
Approval requirement	No
Expected outcome	Agent reports the total_count of Pending tasks from the tool result.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Multi-Tool Cases

Requests requiring more than one tool • minimum 8 required • provided: 9

Case 1 MULTI-TOOL

User request	Here are today's meeting notes: [transcript]. Turn the action items into tasks.
Expected tool(s)	extract_meeting_actions -> create_task (one call per action item)
Expected arguments	Step 1: {"transcript": "..."} Step 2 (per item, after approval): {"title": "...", "priority": "...", "due_date": "..."}
Approval requirement	Yes, before the tasks are created (Workflow A)
Expected outcome	Agent extracts action items, shows them as proposed tasks, waits for approval, then creates the approved tasks and returns their IDs.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 2 MULTI-TOOL

User request	Build me a work plan for today, but first show me what's overdue.
Expected tool(s)	list_tasks -> generate_work_plan
Expected arguments	Step 1: {"status": "Pending", "due_before": "<today>"} Step 2: {"available_hours": <n>, "date": "<today>"}
Approval requirement	No
Expected outcome	Agent first lists overdue tasks, then produces an ordered schedule that accounts for them.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 3 MULTI-TOOL	
User request	Give me a weekly review.
Expected tool(s)	list_tasks (multiple filters) -> internal report generation
Expected arguments	<code>calls: {"due_after":"<week_start>","due_before":"<week_end>"}, {"status":"Completed",...}, {"status":"Blocked",...}</code>
Approval requirement	No
Expected outcome	Agent reports completed vs. overdue counts, lists blocked items, and recommends next week's priorities (Workflow C).
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 4 MULTI-TOOL	
User request	Find my notes about the launch review, and create a task to follow up on the top one.
Expected tool(s)	search_notes -> create_task
Expected arguments	<code>Step 1: {"query":"launch review"} Step 2: {"title":"Follow up on <top note title>","priority":"Medium"}</code>
Approval requirement	No (single task from the follow-up)
Expected outcome	Agent searches notes, picks the top match, and creates a follow-up task referencing it.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 5 MULTI-TOOL	
User request	Show my high-priority tasks, then mark the second one complete.
Expected tool(s)	list_tasks -> complete_task
Expected arguments	Step 1: {"priority":"High"} Step 2: {"task_id":"<2nd result task_id>"}
Approval requirement	Yes, before Step 2 executes
Expected outcome	Agent resolves "the second one" from session memory correctly, shows an approval card for that specific task, and completes it only after approval.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 6 MULTI-TOOL	
User request	Summarize this transcript into action items and set a reminder for the earliest deadline.
Expected tool(s)	extract_meeting_actions -> create_reminder
Expected arguments	Step 1: {"transcript":"..."} Step 2: {"message":"...", "remind_at":"<earliest deadline>"}
Approval requirement	Yes, before the reminder is created
Expected outcome	Agent extracts items, identifies the earliest deadline, then requests approval before scheduling the reminder.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 7 MULTI-TOOL	
User request	Show me tasks due this week and update the launch task's status to Blocked.
Expected tool(s)	list_tasks -> update_task
Expected arguments	Step 1: {"due_after":"<week_start>","due_before":"<week_end>"} Step 2: {"task_id":"<launch task id>","status":"Blocked"}
Approval requirement	Yes, before Step 2 executes
Expected outcome	Agent lists the week's tasks, identifies the launch task, and requests approval before changing its status.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 8 MULTI-TOOL	
User request	Plan my day using today's task list and then save a note summarizing the plan.
Expected tool(s)	generate_work_plan -> save_note
Expected arguments	Step 1: {"available_hours":<n>,"date":"<today>"} Step 2: {"title":"Plan for <today>","content":"<summary>","category":"Planning"}
Approval requirement	No
Expected outcome	Agent generates the plan, then saves a note summarizing it without needing approval (notes are not on the approval list).
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 9 MULTI-TOOL	
User request	Search notes about the Q3 launch, then draft a follow-up email based on what you find.
Expected tool(s)	search_notes -> draft_followup_email
Expected arguments	Step 1: {"query":"Q3 launch"} Step 2: {"recipient_context":"...", "purpose":"...", "key_points":["..."]}
Approval requirement	No for drafting; Yes before it is treated as sent
Expected outcome	Agent searches notes, uses the top result's content as context, and returns a drafted email.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Approval Cases

Requests involving write operations • minimum 5 required • provided: 6

Case 1 APPROVAL	
User request	Update the priority of the launch task to Critical.
Expected tool(s)	update_task
Expected arguments	{"task_id":"<launch task id>","priority":"Critical"}
Approval requirement	Yes
Expected outcome	Approval card shows tool name, task ID, and the field change; task is updated only after Approve.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 2 APPROVAL	
User request	Mark "Interview 3 power users" as complete.
Expected tool(s)	complete_task
Expected arguments	{"task_id":"<resolved id>"}
Approval requirement	Yes
Expected outcome	Approval card shown before the status changes to Completed.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 3 APPROVAL	
User request	Create these three tasks from the meeting notes.
Expected tool(s)	create_task (batch of 3)
Expected arguments	Three separate {"title":..., "priority":..., "due_date":...} objects, one per proposed task
Approval requirement	Yes (batch task creation)
Expected outcome	A single approval card lists all three proposed tasks; all three are created only after Approve, none created on Reject.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 4 APPROVAL	
User request	Set a reminder for the client call tomorrow at 3pm.
Expected tool(s)	create_reminder
Expected arguments	<code>{"message": "Client call", "remind_at": "<tomorrow>T15:00:00"}</code>
Approval requirement	Yes
Expected outcome	Approval card shown; reminder only scheduled after approval.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 5 APPROVAL	
User request	Send the drafted follow-up email to the investor.
Expected tool(s)	draft_followup_email (send/simulate step)
Expected arguments	<code>{"subject": "...", "body": "..."} – confirmation to simulate sending</code>
Approval requirement	Yes
Expected outcome	Agent requests explicit approval before simulating the send; nothing is sent if rejected.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 6 APPROVAL	
User request	Change the status of the onboarding task to Cancelled.
Expected tool(s)	update_task
Expected arguments	<code>{"task_id": "<onboarding task id>", "status": "Cancelled"}</code>
Approval requirement	Yes
Expected outcome	Approval card shown before the status changes to Cancelled.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Failure and Edge Cases

Ambiguous, invalid, or unsupported requests • minimum 4 required • provided: 6

Case 1 EDGE	
User request	(empty message submitted)
Expected tool(s)	None
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent shows a clear message asking the user to enter a request; no tool is called; error handled gracefully (empty_user_input).
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 2 EDGE	
User request	Mark task TSK-9999 as complete.
Expected tool(s)	complete_task (attempted)
Expected arguments	<code>{"task_id": "TSK-9999"}</code>
Approval requirement	Yes (approval requested before the attempt)
Expected outcome	Tool returns a not_found error; agent shows an understandable message instead of a stack trace; no status is changed.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 3 EDGE	
User request	Do my taxes for me.
Expected tool(s)	None — unsupported request
Expected arguments	N/A
Approval requirement	No
Expected outcome	Agent explains this is outside what Trace can do; no tool is called.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 4 EDGE	
User request	Update it.
Expected tool(s)	None yet — clarification needed
Expected arguments	N/A
Approval requirement	No (until clarified)
Expected outcome	Agent asks which task "it" refers to, since nothing was referenced earlier in the session; no tool is called until clarified.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 5 EDGE	
User request	Create a task with priority "Super Urgent".
Expected tool(s)	create_task (attempted)
Expected arguments	<code>{"title":"...", "priority":"Super Urgent"}</code>
Approval requirement	No (fails validation before reaching approval)
Expected outcome	Tool call is rejected with invalid_arguments (priority not in the allowed set); agent asks the user to pick a valid priority.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Case 6 EDGE	
User request	Complete task TSK-9999. (repeated 3 times in a row after each failure)
Expected tool(s)	complete_task (attempted, repeatedly)
Expected arguments	<code>{"task_id":"TSK-9999"}</code>
Approval requirement	Yes (each attempt)
Expected outcome	After the max-retries limit is hit, the agent stops retrying, reports that the task cannot be found, and does not loop indefinitely.
Actual outcome	<i>(fill in after running this request against the live app)</i>
Pass / Fail	<i>(fill in: Pass or Fail)</i>
Notes	<i>(fill in: any observations, discrepancies, or follow-ups)</i>

Evaluation Metrics

Complete the “Measured value” column after running all 36 test cases

Metric	Formula	Initial target	Measured value
Tool Selection Accuracy	Test cases where the correct tool (or correct “no tool” decision) was chosen ÷ total test cases × 100	≥ 85%	<i>(fill in after running all 36 test cases)</i>
Argument Accuracy	Tool calls made with fully correct arguments ÷ total tool calls made × 100	≥ 80%	<i>(fill in after running all 36 test cases)</i>
Task Completion Rate	Requests that reached the expected outcome successfully ÷ total test cases × 100	≥ 80%	<i>(fill in after running all 36 test cases)</i>
Approval Compliance	Sensitive actions where approval was correctly requested ÷ total sensitive-action test cases × 100	100%	<i>(fill in after running all 36 test cases)</i>
Invalid Action Rate	Runs where the agent attempted an incorrect or unsupported action ÷ total test cases × 100	< 10%	<i>(fill in after running all 36 test cases)</i>
Average Response Time	Sum of each run's total duration (from the Execution Log) ÷ number of runs	No fixed target — record as a baseline	<i>(fill in after running all 36 test cases)</i>
Recovery Rate	Tool failures where the agent produced a graceful, understandable outcome ÷ total tool failures observed × 100	No fixed target — record as a baseline	<i>(fill in after running all 36 test cases)</i>

Approval Compliance and Invalid Action Rate should be checked with particular care: any sensitive action (create multiple tasks, update task, complete task, create reminder, send/simulate email) that executed without showing an approval card first is an automatic Approval Compliance failure for that case, regardless of whether the underlying action itself was correct.